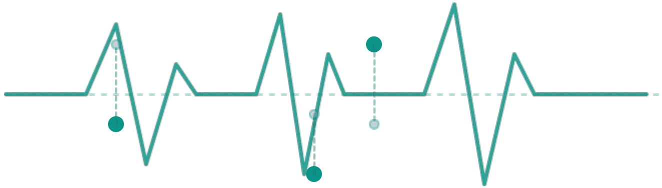
PARALLAX 2026 • PS-S03

NetGuard

Adaptive Diagnostics & Incident Automation



Communication & Networking

ENGINE SPECIFICATION

Automated Wi-Fi Diagnostic & ISP Ticketing Engine

TEAM MEMBERS

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The Problem: Wi-Fi Failures Are Hard to Diagnose

USER SIDE

- Transient Wi-Fi disconnections
- Browsing latency and inconsistent performance
- Non-technical users cannot identify the cause
- Problems may disappear before support investigates

THE GAP
Missing Connection

ISP SIDE

- Limited client-side visibility
- Insufficient diagnostic context
- Vague support requests
- Manual troubleshooting increases resolution time

THE CRITICAL BREAKDOWN

"The ISP receives the complaint — but not the evidence."

OUR GOAL

Automatically capture, investigate and communicate the evidence behind a network incident.

Our Objective

01

Monitor

Continuously observe connection health and Wi-Fi telemetry.

02

Detect

Identify abnormal behaviour using thresholds and recent baselines.

03

Diagnose

Trigger relevant diagnostic tests and investigate the incident.

04

Explain

Provide probable cause, supporting evidence and confidence.

05

Support

Convert the incident into structured information for the ISP workflow.

"From a vague connectivity complaint to an evidence-backed incident."

Literature Study & Research Gap

PAPER 1

RAID: Root Cause Anomaly Identification

Source: ORBilu / ECML PKDD 2025

Objective

Diagnose Wi-Fi performance issues using time-series network KPIs. Combine anomaly detection and lightweight classification for root-cause diagnosis.

Gap relevant to NetGuard

Focuses on Wi-Fi impairment diagnosis rather than an integrated client-to-ISP support workflow.

PAPER 2

Towards a Playground to Democratize AI

Source: arXiv, 2025 / ACM SIGCOMM NGNO

Objective

Provide a standardized and reproducible environment for evaluating AI agents in network troubleshooting.

Gap relevant to NetGuard

Focuses on experimentation and benchmarking rather than continuous client-side telemetry and automated support-case generation.

THE CRITICAL SYNTHESIS

"Research Gap: Continuous client-side monitoring + adaptive diagnostics + evidence-based fault localization + ISP support workflow."

Proposed Solution: NetGuard

STAGE 01

Telemetry

- Latency & Packet loss
- DNS & HTTP reachability
- RSSI & BSSID
- Wi-Fi band & channel
- PHY rate

→

STAGE 02

Diagnostics

- Detect anomalies
- Select dynamic tests
- Gather evidence

→

STAGE 03

Fault Localization

- Correlate telemetry
- Identify fault domain
- Calculate confidence
- Generate explanation

→

STAGE 04

Automation

- Incident records
- Structured payload
- Support workflow

THE ARCHITECTURAL PRINCIPLE

"NetGuard continuously observes the connection and investigates only when the evidence indicates a problem."

Our Novelty

“From detecting a network problem to building evidence for its cause.”

Conventional Troubleshooting

User reports: “Internet is slow”

- Manual diagnostic checklists
- Limited client-side visibility
- Repetitive & exhausting troubleshooting cycles

NetGuard

- Continuous background telemetry
- Automatic, zero-touch anomaly detection
- Adaptive diagnostic selection & test execution
- Evidence-based fault localization (non-opaque)
- Confidence-backed structured support cases

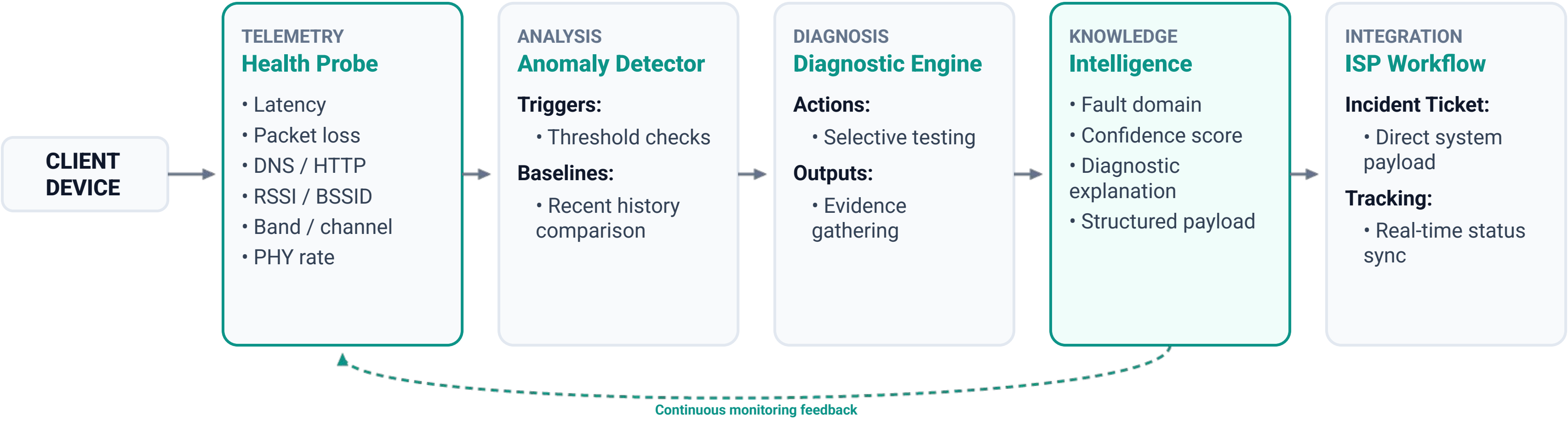
THE NOVELTY INTEGRATION

“Our novelty is the integration of adaptive diagnostics and evidence-based fault localization into a client-to-ISP workflow.”

Feedback Loop

MONITOR → DIAGNOSE → LEARN FROM EVIDENCE

System Architecture



NetGuard: End-to-End Workflow

01

MONITOR
Lightweight probes continuously measure network health.

02

DETECT
Identify abnormal behaviour.

03

CAPTURE CONTEXT
Collect Wi-Fi and device telemetry.

04

INVESTIGATE
Select relevant diagnostic tests based on observed anomaly.

05

LOCALIZE
Correlate evidence to identify probable fault domain.

06

EXPLAIN
Generate supporting evidence and confidence.

07

SUPPORT
Store incident & support info.

DECISION MATRIX

NORMAL → Continue monitoring

ANOMALY → Investigate → Explain → Record → Support

Evidence-Based Fault Localization

Telemetry → Evidence Correlation → Probable Fault Domain → Confidence

LOCAL WI-FI

- RSSI degradation
- Packet loss
- Wi-Fi link behaviour

GATEWAY / LAN

- Increased latency
- Abnormal local network behaviour

UPSTREAM / INTERNET

- Persistent latency or connectivity degradation despite healthy local Wi-Fi indicators

EVIDENCE CHAIN A

RSSI ↓ + Packet Loss ↑
→ **Local Wi-Fi** becomes more likely

EVIDENCE CHAIN B

Latency ↑ + Stable Wi-Fi indicators
→ **Gateway / upstream** becomes more likely

SYSTEM DESIGN NOTE

"Diagnosis is presented as a probable cause supported by measurable evidence, not an unexplained black-box decision."

Implementation: NetGuard Diagnostic Engine

01
runProbe.ts
Network health probing active loops

02
wifiTelemetry.ts
Wi-Fi/RF metrics stream collector

03
anomalyDetector.ts
Real-time behavior drift identifier

04
selectNextTests.ts
Adaptive test dynamic trigger selection

05
flightRecorder.ts
Telemetry ring-buffer incident recorder

06
scoreConfidence.ts
Diagnostic outcome accuracy estimator

07
generateExplanation.ts
Human-readable diagnostic report parser

08
supabaseClient.ts
Cloud backend storage synchronizer

IMPLEMENTATION DETAIL
"The frontend presents the result; the modular diagnostic layer performs the monitoring and incident intelligence."

NetGuard Client Dashboard

01
Real-time connection status

03
RSSI / Wi-Fi band / channel

02
Live network telemetry

04
Live telemetry trends

05
Incident history with probable cause, confidence and supporting evidence

“From raw network measurements to an interpretable incident view.”

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From Network Incident to Support Case

01
RAW TELEMETRY
Continuous network health probing metrics

→

02
ANOMALY
Identified behavior drift & threshold failures

→

03
DIAGNOSTIC EVIDENCE
Collected telemetry & active test results

→

04
PROBABLE CAUSE
Correlated fault domain with confidence score

→

05
ISP / SUPPORT CONSOLE
Direct evidence sync to operator dashboard

STRUCTURED INCIDENT

Complete diagnostic payload packaged for support integration.

Incident ID: INC-2026-99214

Timestamp: 2026-08-18 08:42:15 UTC

Fault domain: Local Wi-Fi (RF Congestion)

Probable cause: Channel 36 over-saturation

Confidence: 92% (High)

Supporting evidence: BSSID probe failures + high retry rate

Relevant telemetry: RSSI -78 dBm, Packet Loss 14%

Incident status: READY FOR ISP

"Instead of asking the ISP to start diagnosis from zero, NetGuard transfers the evidence with the incident."

Impact & Future Scope

01 / KEY IMPACT

• **Faster fault localization**: Reduces time-to-resolution for complex network issues.

• **Better ISP context**: Eliminates blind spots by providing detailed telemetry directly to providers.

• **Reduced troubleshooting**: Minimizes manual diagnostic effort from end-users.

• **Transient visibility**: Captures intermittent Wi-Fi drops that disappear during active support sessions.

• **Clearer interactions**: Empowers users with logical, evidence-backed support details.

02 / FUTURE SCOPE

• **Background Agents**: Native mobile and desktop clients to continuous monitor health.

• **Ticketing APIs**: Direct, automated evidence injection into provider-side support desks.

• **Distributed Testing**: High-density diagnostic nodes to isolate local vs global routing paths.

• **Telemetry Datasets**: Expanded anonymized signal repository to fuel trend forecasting.

• **Learned Diagnosis**: Advanced predictive models trained to pinpoint failure patterns.

“NetGuard turns an intermittent Wi-Fi complaint into a measurable, explainable and actionable incident.”